Listing 6.5 Edabit challenge: Plant the Grass

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Plant the Grass by AniXDownLoe

    You will be given a matrix representing a field g
and two numbers x, y coordinate.

    There are three types of possible characters in the matrix:

        x representing a rock.
        o representing a dirt space.
        + representing a grassed space.

    You have to simulate grass growing from the position (x, y).
    Grass can grow in all four directions (up, left, right, down).
    Grass can only grow on dirt spaces and can't go past rocks.

    Return the simulated matrix.
    Examples

    simulate_grass([
    "xxxxxxx",
    "xooooox",
    "xxxxoox"
    "xoooxxx"
    "xxxxxxx"
    ], 1, 1) → [
    "xxxxxxx",
    "x++++++x",
    "xxxx++x"
    "xoooxxx"
    "xxxxxxx"
    ]

    Notes

    There will always be rocks on the perimeter
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You can use any challenge or coding exercise you want, but here are a few things to consider:

- The challenge should be testable with quantifiable assertions (pass/fail).
- Avoid opening windows when asking for a game, building a website, or using another interface. At some point, testing full interfaces will be possible, but for now, it’s just text output.
- Avoid long-running challenges, at least initially. Start by keeping the challenges concise and short lived.

Along with any challenge, you’ll also want a set of tests or assertions to confirm the solution works. On Edabit, a challenge typically provides a comprehensive set of tests. The following listing shows the additional tests provided with the challenge.